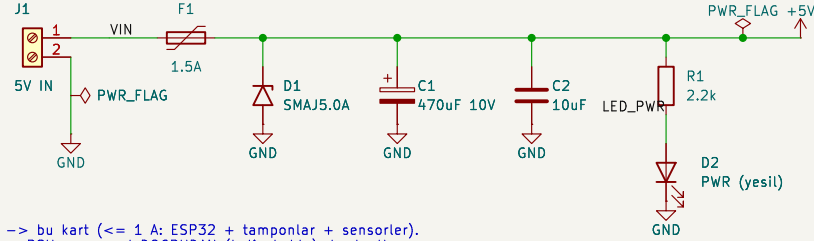
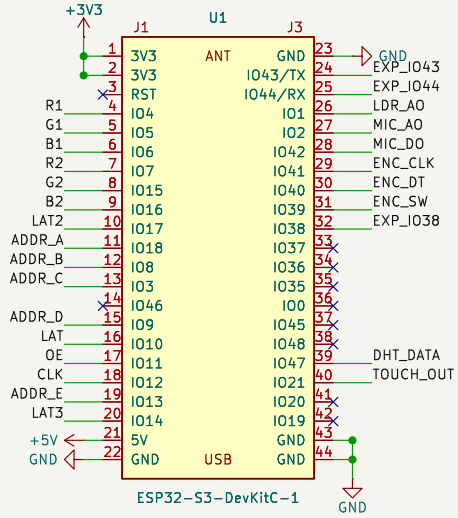


1 5 V GIRIS / KORUMA



PSU 5 V -> bu kart (<= 1 A: ESP32 + tamponlar + sensorler).
Panel gucu PSU -> panel DOGRUDAN (kalın kablo), bu karttan geçmez.
Ters kutup / asiri gerilim: D1 iletir -> F1 (PTC) acar, kendini sıfırlar.

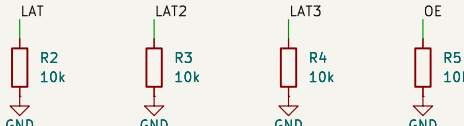
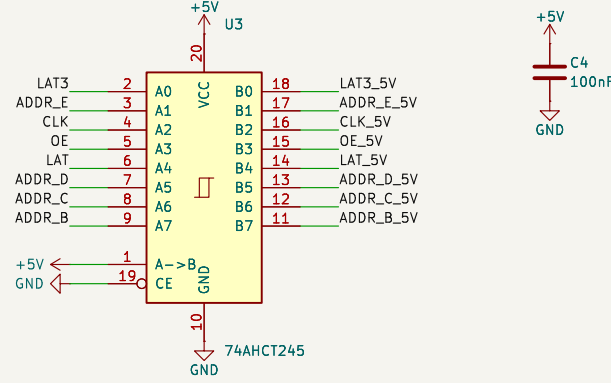
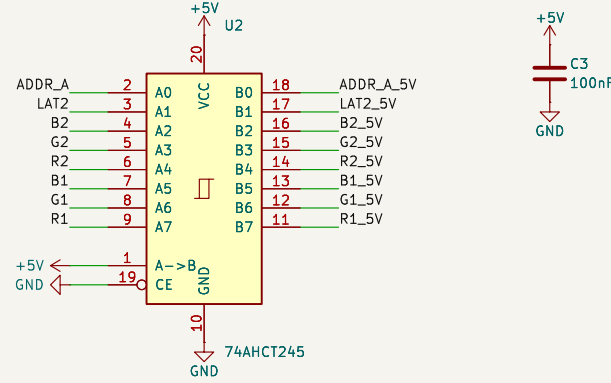
2 ESP32-S3-DevKitC-1 (soket)



ESP32-S3-DevKitC-1

Sag soket iki sirali: 0.9in (resmi Espressif) / 1.0in (N16R8 klonlari).
Anten (üst) kart kenarından tasar; USB (alt) kart kenarında.
Kullanılmayan: IO0/45/46 strapping, IO35-37 OPI PSRAM, IO19/20 USB, IO48 (klon RGB LED).

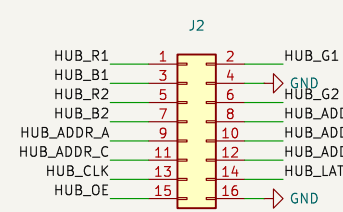
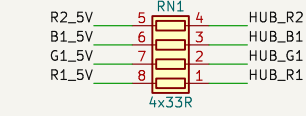
3 SEVIYE DONUSTURUCU 3.3 V -> 5 V (2x 74AHCT245)



Kanal sırası DevKit J1 pin sırasını izler (PCB: kesisimsiz).
Pull-down (R2-R5): acilista LAT/LAT2/LAT3/OE dusuk -> panel karanlik kalir.

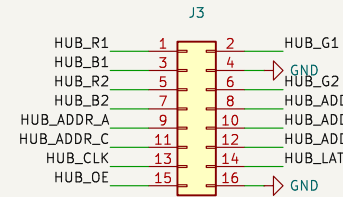
4 HUB75E CIKISLARI (seri 33R, 3 panel)

PANEL 1: LAT = IO10 (P4 ve P1.86)



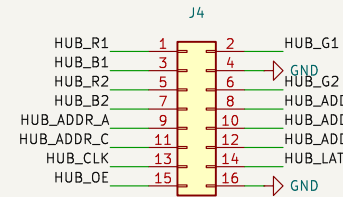
HUB75E PANEL 1

PANEL 2: LAT2 = IO17 (yalnız çoklu P1.86)



HUB75E PANEL 2

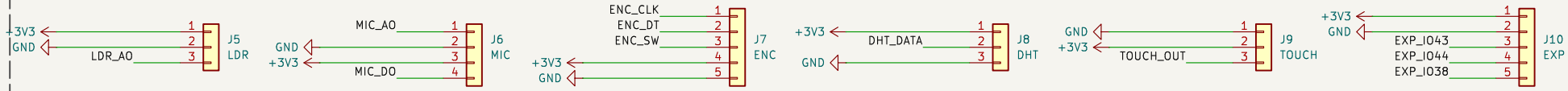
PANEL 3: LAT3 = IO14 (yalnız çoklu P1.86)



HUB75E PANEL 3

HUB75E: 1 R1 2 G1 3 B1 4 GND 5 R2 6 G2 7 B2 8 E
9 A 10 B 11 C 12 D 13 CLK 14 LAT 15 OE 16 GND
Çoklu P1.86: tüm hatlar paralel, LAT panel başına ayrı.

5 SENSOR / KONTROL BASLIKLARI (3.3 V)



LDR Sensor Board (JSUMO 16067)
AO -> IO1 (ADC1_CH0)

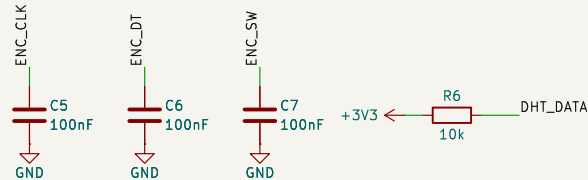
Microphone Sound Sensor (JSUMO 15771)
AO -> IO2, DO -> IO42

KY-040 Mechanic Encoder
CLK/DT/SW -> IO41/IO40/IO39

DHT11 Temperature & Humidity (JSUMO 15579)
DATA -> IO47

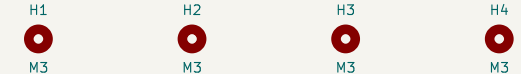
TTP223B Digital Touch (JSUMO 17538)
OUT -> IO21

Genisleme (UART0/leride I2C)
IO43, IO44, IO38 (v1.1 RGB LED)



Pin sırası yaygın modul baskısını izler; PCB üzerinde her pin etiketli.
Bağlanmamış sensor zararsız (firmware: dokunmatik pull-down, enkoder pull-up).

6 MEKANİK



hardware/kicad/gen_carrier.py ile üretildi - elle düzenleme yerine üreticiyi değiştir
Pinler: include/app_constants.hpp, include/panel_sm16380.h, include/sensors.h
Seviye donusturucu: 2x 74AHCT245 (3.3 V -> 5 V), seri 33R, LAT/OE pull-down
MagPanel (github.com/Kem4tk/magpanel)

Sheet: /
File: magpanel-carrier.kicad_sch

Title: MagPanel Carrier - ESP32-S3 + 3x HUB75E + sensorler

Size: A3 Date: 2026-10-01

KiCad E.D.A. 8.0.9

Rev: 1.0

Id: 1/1